



Dentin Hypersensitivity

Definition

- Dentin hypersensitivity is characterized by short, sharp pain arising from exposed dentin in response to stimuli typically thermal, evaporative, tactile, osmotic or chemical and which cannot be ascribed to any other form of dental defect or pathology.
 - » Dowell, 1985
 - » Orchardson and Collins, 1987

- Two process must therefore occur for dentin hypersensitivity to manifest
 - Localization – Dentin must be exposed by etiologic agents
 - Initiation – sensitivity must be induced by tubule exposure

Prevalence

- The age range – early teenage to 70 years
- Peak incidence – 3rd and 4th decades of life
- Gender differences – females more than males

Distribution

- Dentin hypersensitivity can occur anywhere in the oral cavity and affect any tooth,
- Areas of higher predilection are:
 - Buccal cervical area of teeth is the site of predilection for dentin hypersensitivity
 - Other areas of predilection include areas of gingival recession and where enamel is thin.
- Most commonly affected are the canines and 1st premolars, then incisors and 2nd premolars and least often molars.
- Tooth sensitivity is more common on the left side than on the right side

Etiology of hypersensitivity

- Dentin hypersensitivity can occur by dentin exposure which can be by 2 methods
 - Loss of enamel due to tooth wear
 - Conditions leading to recession of gingiva and subsequent loss of cementum
- One of the iatrogenic cause of tooth sensitivity is vital tooth bleaching

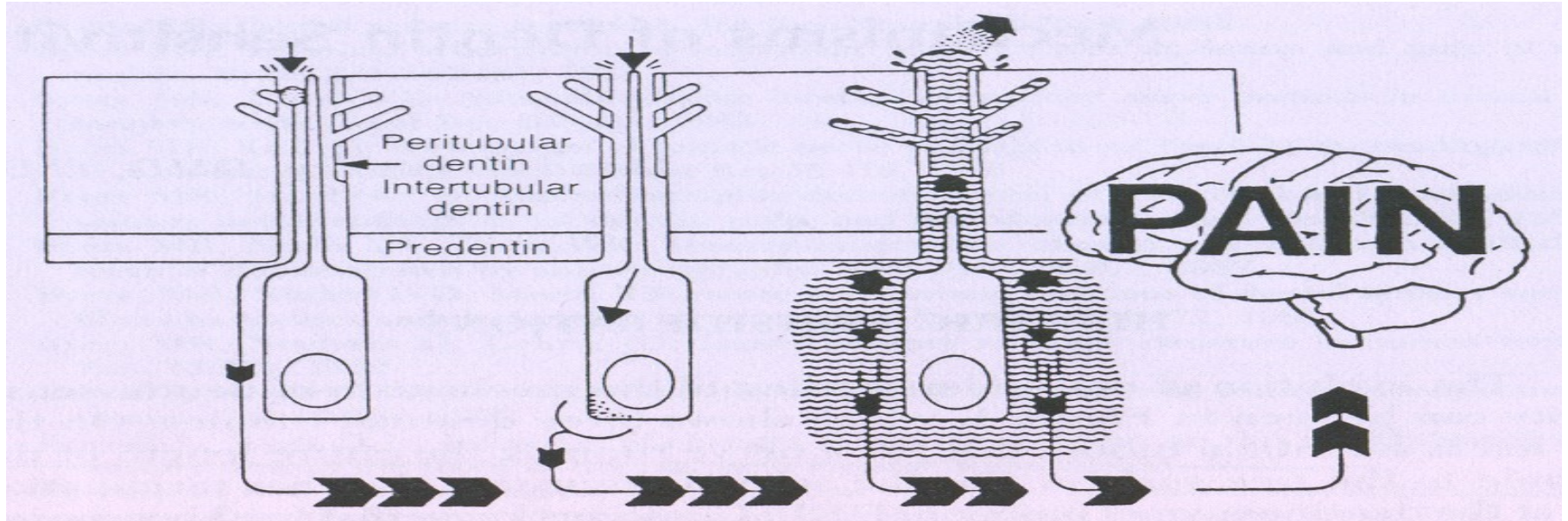
Predisposing factors for gingival recession

- Aging is a physiologic etiologic agent
- Anatomy of the alveolar bone
- Tooth anatomy and tooth position
- Excessive tooth brushing to maintain good oral hygiene
- Non surgical and surgical periodontal procedures
- Acute gingival infections and periodontal diseases.
- Some conditions like
 - Fraenal pull
 - Occlusal trauma
 - Emotional stress

Conditions causing loss of enamel

- Tooth wear (Attrition, Abrasion, Erosion & Abfraction)
- Fracture of tooth
- Developmental lesions that cause enamel loss.

Theories of dentin hypersensitivity



Rapp and colleagues, 1968

Gysi's concept, 1900

Anderson and Brannstrom independently 1972

DENTIN CONSIDERATIONS

- There are two mechanisms responsible for the permeation of substances across dentin:
 - diffusion and
 - convection
- **Diffusion** is the process by which substances are transported from an area of high concentration to an area of low concentration.
 - there is no bulk fluid movement but only molecular translocation.
 - driving force is a concentration gradient or chemical potential energy
- **Convective transport or filtration** - bulk fluid movement occurs from an area of high hydrostatic pressure to an area of low hydrostatic pressure.

- According to the hydrodynamic theory of dentin sensitivity, it is fluid movement that is responsible for transducing a variety of physical stimuli into electrical nerve activity.
- This type of fluid movement can be quantitated by measuring the hydraulic conductance of dentin which is the reciprocal of resistance.
- dentin with a high conductance has a low resistance

Poiseuille Hagen Equation

- The variables regulating hydraulic conductance are
 - The length of the tubules (that is, dentin thickness) - L
 - The applied pressure - ΔP
 - The viscosity of the fluid, and - η
 - the radius of the tubules raised to the fourth power - r^4
 - The number of tubules per unit surface area - N

$$\text{Fluid flow } Q = \frac{\pi \Delta P r^4 N}{8 \eta L}$$

- Methods of evaluating pain associated with dentin hypersensitivity

- Management of dentin hypersensitivity

Potential treatment modalities

- **Home use products**

- **In-office products**

Home use products

- The products include agents such as potassium salts, strontium salts and fluoride salts in toothpaste, mouthwash and gel formulations.
- These reduce the symptoms by
 - Occluding dentine tubules and/or
 - Intercepting the neural response by chemical intervention
- Advantages
 - Readily and widely available
 - Cost effective
 - Simple to use
 - Non invasive

- The effect of self-applied treatments in reducing the symptoms is not generally noticed immediately, it may take some time.
- The effect may also have a short lifetime following the cessation of treatment.

Various agents used in dentifrices

- 10% Strontium chloride hexahydrate
- 5% Potassium nitrate
- Dibasic sodium citrate
- Fluorides

In-office products

- In-office treatment of dentine hypersensitivity should be performed in all cases of severe hypersensitivity
- The treatment options are often dictated by the clinical condition.
- All in-office treatments are designed to decrease the hydraulic conductance of dentin by partial or complete tubule occlusion
- The most difficult sites to treat are those where there is no loss of tooth structure

- It is crucial that the exact region of hypersensitivity be identified.
- Air blasts are useful as a screening tool, they cannot identify the exact site
- Surfaces should be gently evaluated with a dental explorer, in a raster-like pattern across the exposed surface
- All treatments should be done on unanesthetized patients

Treatment planning

- The clinical use of these agents will depend on
 - the degree of sensitivity,
 - the number of teeth involved,
 - the amount of tooth structure lost and
 - the amount of treatment time that would be required
- Initially the treatment is restricted to non invasive techniques like application on varnishes / oxlate solutions are used
- If relief is not felt complex treatment tried such as the use of a Glass ionomers or other adhesive material is advocated.

Criteria for the Selection of Desensitizing Agents

- Provides immediate and lasting relief of pain
- Easy to apply
- Well tolerated by patients
- Not injurious to the pulp
- Not discolor the tooth
- Relatively inexpensive

In Office Treatments for Dentinal Hypersensitivity

- Cavity varnishes
- Anti-inflammatory agents
- Treatments that partially obturate dentinal tubules
 - Burnishing of dentin
 - Silver nitrate
 - Zinc chloride-potassium ferrocyanide
 - Calcium compounds
 - Calcium hydroxide
 - Dibasic calcium phosphate
 - Fluoride compounds
 - Sodium fluoride
 - Sodium silicofluoride
 - Stannous fluoride
 - Primers containing HEMA
 - 5% glutaraldehyde, 35% HEMA in water
 - 35% HEMA in water
 - Strontium chloride
 - Potassium oxalate
 - Iontophoresis
 - Lasers

In Office Treatments for Dentinal Hypersensitivity

- Restorative methods
 - Conventional glass ionomer cements
 - Resin-reinforced glass ionomers / compomers
 - Adhesive resin bonding systems

Cavity Varnishes

- Dentin often becomes insensitive when open tubules are covered with a thin film of varnish
- An effective means of providing temporary relief
- For more sustained relief, a fluoride containing varnish, Duraflor can be applied

Corticosteroids

- Use of corticosteroids is based on the assumption that hypersensitivity is linked to pulpal inflammation
- Corticosteroids caused complete obliteration of tubules, thus decreasing dentin permeability.
 - » Furseth, 1968
- Mosteller (1962) used liner consisting of
 - 1 % prednisolone in combination with
 - 25 % para-chlorophenol,
 - 25 % M-cresyl acetate,
 - 50 % gum camphor.

Burnishing of dentin

- Results in the formation of a smear layer that partially occludes the dentinal tubules
- Burnishing created a partial smear layer that reduced fluid movement across dentin by 50 to 80 per cent

» Pashley, 1987

Formation of Insoluble Precipitants to Block Tubules

- Certain soluble salts react with ions in tooth structure to form crystals on the surface of the dentin
- AgNO_3 is a time honored desensitizing agent.
- It has ability to precipitate protein constituents of odontoblast processes thereby partially blocking the tubules
- AgNO_3 followed by 10 per cent formalin reduced hydraulic conductance by 59.2 per cent whereas AgNO_3 alone reduced it by 47.4 per cent.

- Other protein precipitants such as zinc chloride and phenol can also be used
- Gottlieb developed zinc chloride-potassium ferrocyanide impregnation method
 - 40 % solution of aqueous zinc chloride for 1 minute, followed by a 20 % aqueous solution of potassium ferrocyanide rubbed on dentin surface.
 - Scanning electron micrographs of this precipitate have revealed a highly crystalline deposit covering the dentin surface.

CALCIUM COMPOUNDS

- The exact mechanism of action is unknown, but evidence suggests that it
 - May block dentinal tubules or promote peritubular dentin formation.
 - Increasing the concentration of calcium ions around nerve fibers can result in decreased nerve excitability, and thus suppressing nerve activity.
 - May stimulate reparative dentine formation
- Mjor observed increased radiodensity in the $\text{Ca}(\text{OH})_2$ covered dentin.
- Using SEM, Brannstrom observed a variable constriction of the dentinal tubules in the majority of teeth treated with $\text{Ca}(\text{OH})_2$

FLUORIDE COMPOUNDS

- Lukomsky (1941) - first to propose sodium fluoride as a desensitizing agent
- NaF reacts with calcium and phosphate ions of dentinal fluid leading to precipitation of CaF_2 crystals, thus reducing the functional radius of the dentinal tubules
- NaF has less effect on dentin permeability because
 - Size of CaF_2 crystal is very small
 - Fluoride is lost fairly rapidly
- Various forms of fluoride used
 - Acidulated sodium fluoride
 - Sodium silico fluoride
 - Stannous fluoride
 - Fluoride iontophoresis

Strontium Chloride

- Strontium strongly binds to calcified tissues
- Strontium deposits are produced by an exchange with calcium in the dentin resulting in re-crystallization in the form of a strontium apatite complex. These crystals are responsible for reducing the sensitivity.
- Topical application of 10 % SrCl_2 prior to application of 2 % NaF was more effective in decreasing sensitivity than NaF alone

» Gedalia, 1978

Oxalates

- Oxalates have gained great popularity as a desensitizing agent
- They are relatively inexpensive, easy to apply, and well tolerated by patients.
- Oxalate ions react with calcium ions in the dentinal fluid to form insoluble calcium oxalate crystals.
- calcium oxalate are resistant to dietary acids.
- Oxalate compounds developed as desensitizers are
 - 6 % ferric oxalate,
 - 30 % dipotassium oxalate,
 - 3 % monohydrogen-monopotassium oxalate.

Restorative methods

- Conventional glass ionomer cements
 - The cervical lesions were etched with 50% citric acid for 30-45 s, then rinsed and dried prior to placement of the glass ionomer cement.
- Resin-reinforced glass ionomers were also used
- Adhesive resin primers - use of adhesive primers to occlude the open tubules is popular.

Adhesive resin bonding systems

- Resin desensitization is due to the formation of resin tags in the tubules
» (Yoshiyama *et al.* 1993).
- These materials are somewhat technique sensitive and should be used carefully
- The latest generation of adhesive bonding systems are hydrophilic and provide better bonds in wet environments
- The disadvantage of the adhesive systems is that their polymerization is inhibited by atmospheric oxygen to a depth of 10-15µm

Adhesive resin bonding systems

- Javid *et al.* (1987) evaluated a single application of cyanoacrylate to hypersensitive root surfaces, and found a significant immediate reduction in sensitivity
- Orchardson *et al.* (1993) advocated multiple applications (twice weekly for 4 weeks) of a fluoride-containing adhesive bonding system.

Iontophoresis

- Iontophoresis is a term applied to the use of an electrical potential to transfer ions into the body for therapeutic purpose
- The objective of iontophoresis is to drive the topically applied therapeutic agents more deeply into the dentinal tubules.
- Use of iontophoresis of NaF to treat hypersensitive dentine has been advocated by Gangarosa (1983)
- Disadvantages
 - It is a technique sensitive method
 - Apparatus required is expensive
- Pashley found that the use of AgNO_3 or potassium oxalate produced significantly greater reductions in hydraulic conductance than fluoride iontophoresis.

Lasers

- Nd:YAG laser used for 2 min at 10 pulses/s at lower power levels can be used
- These laser treatments were all done at power levels below that required to char dentin
- Mechanism of action is by
 - Coagulation and precipitation of plasma proteins in dentinal fluid - Goodis *et al.* (1994)
 - The thermal energy can alter intradental nerve activity (Orchardson *et al.* 1998)

Conclusion

- Because of improved caries prevention, greater awareness of the importance of preserving all masticatory units, and change in the life style and dietary habits, we might face increased frequency of non carious lesions
- Depending on the degree of tooth wear, restorative treatment can range from placement of bonded composites in a few isolated areas of erosion, to full mouth reconstruction in the case of the devastated dentition.

Conclusion

- Regardless of the type of restorative therapy provided, prevention of the progression of tooth wear should be the basis of management.
- This will increase the likelihood of successful, long-term outcomes of the restorative treatment.
- The patient should be monitored at regular intervals to determine compliance and success of treatment.
- Thus, apart from regular monitoring of the aetiology and development of these lesions, improved measures of prevention and therapy still need to be introduced.